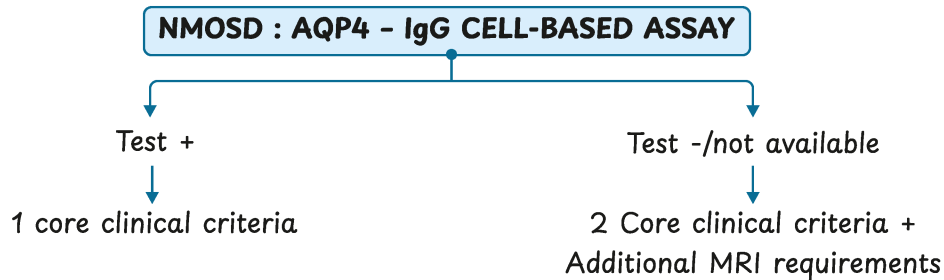




- **V**isual evoked potentials (B/L)
- **L**ETM > 3 contiguous spinal cord segments (involvement of grey matter)
- **E**-

DIAGNOSIS

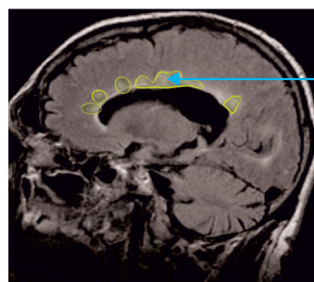
00:10:43



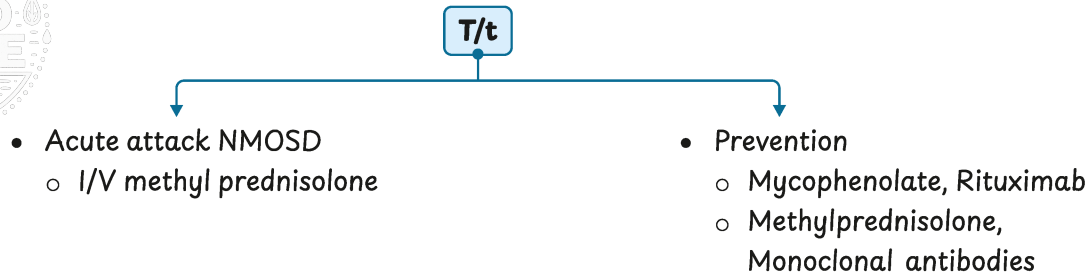
ADDITIONAL MRI REQUIREMENTS

- >1/2 optic nerve involvement/optic chiasma
- LETM - ≥ 3 contiguous segments of spinal cord
- Dorsal medulla involvement - Area postrema involvement
- Peri ependymal brainstem lesion - Cloud like lesions

FEATURES	NMOSD	MULTIPLE SCLEROSIS
ETHNICITY	Asian/ African	Caucasian
CLINICAL FEATURES	B/L optic neuritis (sudden blindness)	U/L visual acuity $\downarrow$, color vision $\downarrow$
MRI BRAIN	Cloud lesion	Dawson fingers
MRI SPINE	LETM	+/-
ASSOCIATED AUTOIMMUNITY	SLE/ Hashimoto thyroiditis	Hypothyroidism
CSF FINDINGS	• $\uparrow$ Neutrophils • $\uparrow$ Eosinophils • Oligo clonal bands < 20%	Oligo clonal bands



Dawson fingers



MONOCLONAL ANTIBODIES

- Eculizumab
 - Binds to C_5 → inhibits cleavage of C_5 (C_{5A} & C_{5B}) → inhibition of activation on complement system
- Inebilizumab
 - Binds to CD_{19} → depletes B cells
- Satralizumab
 - Blocks $IL-6$ receptor → prevents cytokine induced damage



- Q. A 32-year-old female presents to the neurology clinic with visual disturbances including blurred vision along with weakness and sensory changes in her lower limbs. Neurological examination spastic paraparesis. Anti AQP4 antibody is positive. Which is not a feature of this presentation?
- a. More than half the length of the optic nerve involvement showing neuritis
 - b. Dorsal medulla involvement
 - c. Periventricular brain lesions
 - d. Longitudinally extensive transverse myelitis

Ans: (c)



AMYOTROPHIC LATERAL SCLEROSIS AND MOTOR NEURON DISEASE



AMYOTROPHIC LATERAL SCLEROSIS

00:02:23

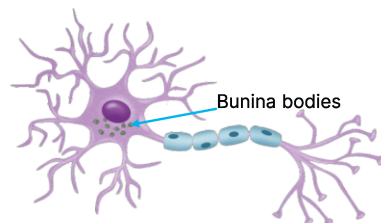
- M/c subtype of motor neuron disease: Amyotrophic Lateral Sclerosis
- Amyotrophic
 - Denervation of muscles → features of LMN disease
 - Destruction of Anterior horn cells
- Lateral sclerosis
 - Fibrillary gliosis of lateral column (involvement of corticospinal pathway)
 - (+) Babinski sign
 - Features of UMN lesion
- Neuro illnesses showing both UMN + LMN lesion manifestations (Buzzwords: Babinski sign with distal areflexia)
 - SADC: Vit B₁₂ deficiency → demyelination
 - In peripheral nerves: Areflexia
 - In corticospinal tract: (+) Babinski sign
 - ALS: Familial
 - Frederick ataxia:
 - Defective Frataxin gene
 - DM-II
 - HCM
 - Brown Sequard syndrome- hemisection of spinal cord

Important Information

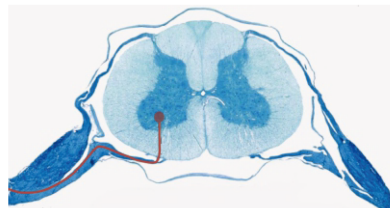
- Polio : Destruction of anterior horn cells

PATHOPHYSIOLOGY

- Damage of anterior horn cells
- Intracytoplasmic bodies → **Bunina bodies**
 - Accumulation of Ubiquitin (Aggregates of TDP-43)



00:08:38

GENE INVOLVED	% CONTRIBUTION TO FALS	INHERITANCE	NOMENCLATURE
SOD1 gene defect • SOD1 gene → codes for the activities of SOD enzyme • SOD enzyme neutralizes the free radicals	20%	AD	ALS-1 
C9 orf 72 gene defect	50%	AD: hexaneucleotide repeats	ALS-FTD

BIO-MARKERS FOR MOTOR NEURON DISEASE

00:13:55

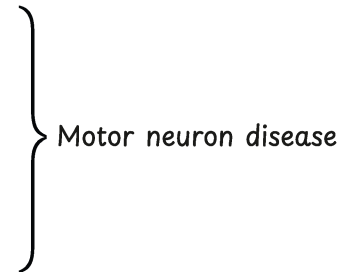
- CSF microcyte chemoattractant protein : ↑
- CSF neurofilament light chains : ↑
- CSF neurofilament heavy chains : ↑

Important Information

- CSF Kappa light free chains : ↑ - Multiple sclerosis
- CSF Aβ42 : ↓ - Alzheimer's disease
- CSF neurofilament light chains : ↑ - ALS

DAMAGE TO

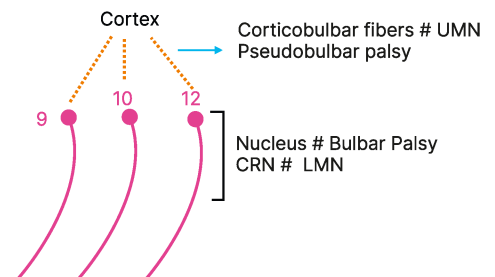
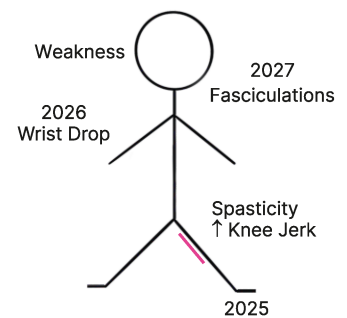
- LMN of the spinal cord, Brainstem, CN nuclei : **LMN**
 - Primary muscular atrophy
 - Bulbar palsy
- UMN of the spinal cord, Brainstem, CN nuclei : **UMN** (Corticobulbar fibers)
 - Primary lateral sclerosis
 - Pseudobulbar palsy
- Combination : **LMNL + UMN** → Amyotrophic lateral sclerosis



CLINICAL FEATURES

- Slow progressive muscle weakness → Asymmetric initially
- Recurrent falls
- Recurrent cramps in gym
- Poor hand grip
- Fasciculations, wasting
- Absent biceps/ triceps jerk
- Bulbar weakness: CN IX, X, XII involved
 - (-) gag reflex
 - (-) cough reflex
- Pseudobulbar palsy
 - Corticobulbar fibers involved: CN IX, X, XII
 - Speech: dysarthria, Donald duck voice
 - Exaggeration of emotions
 - Gag reflex: exaggerated
- **Sensory system not involved**
- **Bladder, bowel & ocular motility preserved**
- Respiratory failure: cause of death
- Death: 3 - 5 years

00:19:44



Muscle weakness



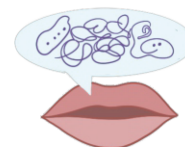
Muscle Cramps



Twitching



Stiff muscles



Speech challenges



Trouble swallowing



Drooling



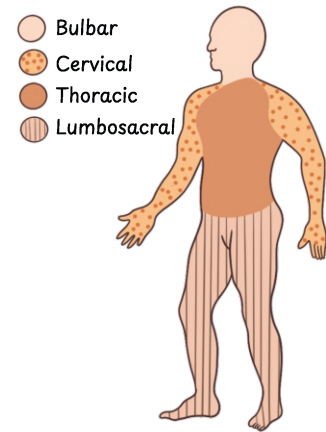
Involuntary emotional expressions



Fatigue

MOTOR NEURON DISEASE

- If 3 out of 4 areas involved:
 - FMA - LMNL
 - PLS - UMNL
 - ASL - LMNL + UMNL
- Involvement of motor neurons:
 - Bulbar: can't eat/ drink/ talk
 - Cervical: arm weakness, fasciculations, wrist drop
 - Thoracic:
 - Lumbosacral: leg weakness, Babinski sign



00:29:24

DIFFERENTIAL DIAGNOSIS

- Tumors of cervical spine
 - MRI spine
- Multifocal motor neuropathy with conduction block
 - IgM- Anti Gm₁
 - LMNL features
- Multiple myeloma
- Lyme disease
- Lead poisoning → Basophilic stippling in RBC

00:31:18

TREATMENT

- Edavarone
- Riluzole
- Phenylbutyrate : tau ursodiol
- None of the medicines are DOC, they only help in delaying the progression of the disease
- Supportive devices
 - Cough assist device
 - Tracheostomy
 - Gastrostomy
 - Speech synthesizers

00:32:21

PAEDIATRIC VARIANT OF PRIMARY MUSCULAR ATROPHY

- Defect in SMN gene (survival motor neuron gene)
- Floppy baby at birth → Werdnig Hoffman syndrome
- New drug: Nusinersen (antisense oligonucleotides)



Q. 20 year old male presents with history of recurrent falls and progressive weakness in hand grip on right side for last one year. On examination he had fasciculations in muscles of upper limb with absent biceps jerk, spastic weakness in right leg with extensor plantars. EMG showed a denervation pattern upper limbs. Which of the following is first D/D?

- a. Becker muscular dystrophy
- b. Charcot Marie tooth disease
- c. Motor neuron disease
- d. Chronic inflammatory demyelinating polyneuropathy

Ans. (c)

Q. Correct about Multifocal Motor Neuropathy with conduction block.

- a. Presents as spastic weakness in upper limbs
- b. Caused by C9orf72 mutation
- c. CSF electrophoresis shows oligoclonal bands
- d. IgM Anti-GM1 antibodies are seen

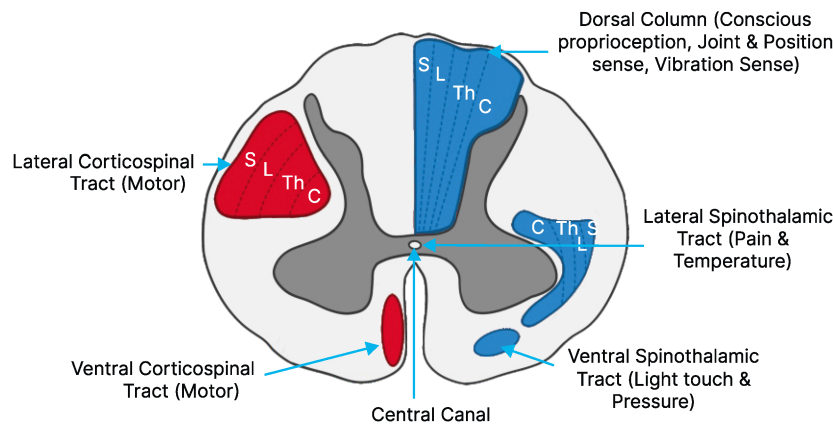
Ans. (d)



SYRINGOMYELIA AND CONUS MEDULLARIS



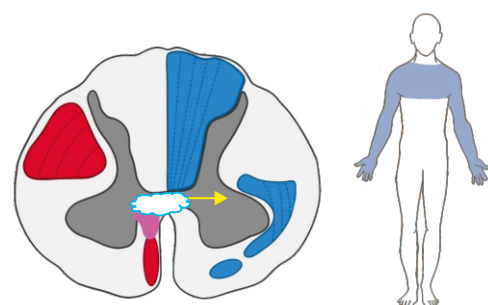
BASICS



- In corticospinal tract,
 - Cervical fibers - most medially present
 - Sacral fibers - most laterally present
 - In case of extradural compression sacral fibers are compressed first
 - In cases of tumors inside (Meningioma/ Ependymoma) → cervical fibers compressed initially → **sacral sparing**
- Lateral spinothalamic tract
 - Carries 2 sensations - **pain and temperature**
 - Cervical fibers - most medially present
 - Sacral fibers - most laterally present
 - In Syringomyelia → cervical fibers involved initially
 - Initial manifestation in arm → **cape** distribution
- Ventral spinothalamic tract
 - Carries 2 sensations - **light touch and pressure**
- Dorsal column
 - Responsible for - conscious proprioception/Joint and position sense and vibration sense
 - Sacral fibers - present most **medially**
 - Cervical fibers - present most laterally

SYRINGOMYELIA

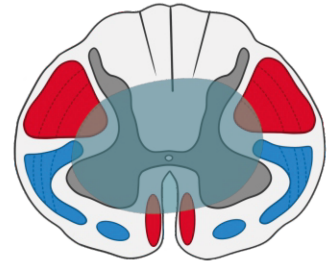
- Dilation of central canal of spinal cord
- Compresses lateral spinothalamic pathway (pain and temperature)
- Compresses cervical fibers → **painless burns**, arms involved (Cape distribution of sensory loss)



00:03:50

CENTRAL CORD SYNDROME

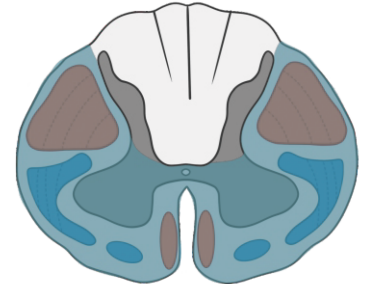
- Syringomyelia - one of causes of central cord syndrome
- Expansion of syrinx → medial fibers involved (cervical fibers) → spastic weakness with brisk reflexes in arm
- Involvement of lateral spinothalamic pathway → loss of pain and temperature sensation in arms



Central cord syndrome
(Area affected)

ANTERIOR SPINAL ARTERY INFARCTION - ANTERIOR CORD SYNDROME

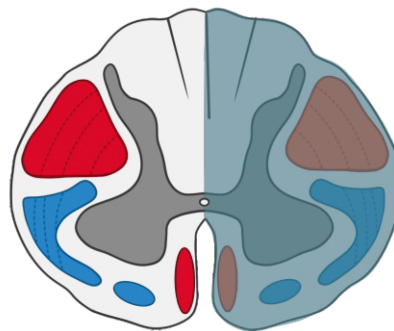
- Anterior spinal artery supplies anterior 2/3 of the spinal cord
- Affected pathways -
 - B/L corticospinal pathways
 - B/L lateral spinothalamic pathways
- **Dorsal column spared** (conscious proprioception, joint & position sense, vibration)



Anterior Cord Syndrome
(Area affected)

BROWN SEQUARD SYNDROME

- Hemisection of spinal cord

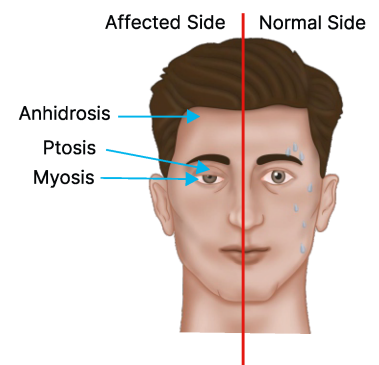


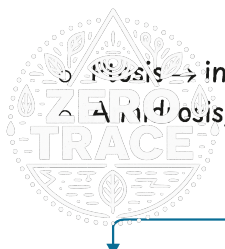
Brown Sequard Syndrome
(Area affected)

FEATURES OF SYRINGOMYELIA (MNEMONIC- BURN CASE)

00:07:07

- **C**avitation in central canal causing
 - Regional dissociated sensory loss
 - Loss of pain and temperature sensation
 - Painless **burns**
 - **S**pared - Crude touch, proprioception, vibration sensation
- **C**ape distribution
- **A**reflexic weakness in shoulder, arm and hands - lower motor neuron lesion at level of lesion
 - Absence of Biceps jerk
 - Intrinsic muscle wasting (↓ hand grip)
 - Multiple levels of spinal cord involved → based on extension of lesion
- **S**yrx formation causing spasticity in legs
 - Involvement of corticospinal tract - loss of inhibitory control
- **S**ensation loss in face
 - CN V nucleus extends up to C₂ cervical cord
- **S**paring of dorsal column
- **HornE**r syndrome
 - Compression of descending sympathetic fibers





• Miosis → involvement of Mullers muscle

• Anhidrosis, Miosis

OTHER FEATURES

Bladder and bowel involvement

Cough induced headache (Arnold Chiari malformation - 1)

- Syringobulbia
 - Bulbar palsy
 - CN XI, X, XII involved
 - Can't talk, can't drink, can't eat

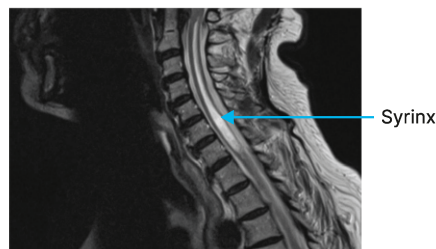
CAUSES OF SYRINGOMELIA

- Arnold Chiari malformation - 1
- Trauma
- Necrotic spinal cord tumor
- Chronic arachnoiditis

Important Information

- Arnold Chiari malformation - 2
 - Associated neural tube defect → myelomeningocele

DIAGNOSIS



00:17:50

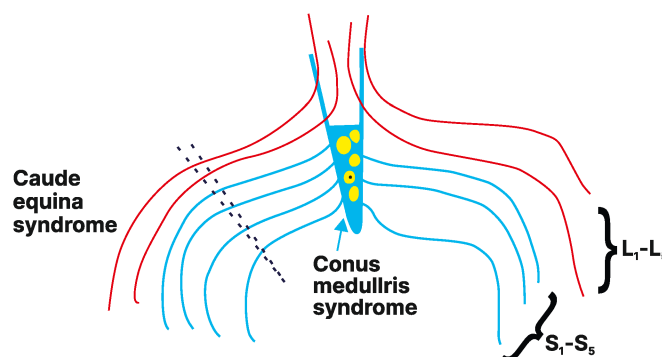
- IOC - MRI -spine

T/t

- Arnold Chiari malformation - 1
 - Hydrocephalus - VP shunt
- Suboccipital Craniectomy - TOC

CONUS MEDULLARIS SYNDROME V/S CAUDA EQUINA SYNDROME

00:19:28

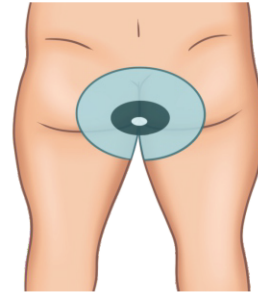


CONUS MEDULLARIS SYNDROME

- Flaccid areflexia
- One of causes → metastasis of prostate cancer
- S1-S5 fibers affected
- Knee jerk - L2, L3, L4 → Preserved
- Ankle jerk - S1, S2 → B/L loss
- Perineum sensation - S2, S3, S4 → Lost
- Bladder/ rectum → Difficulty in starting urination
- Urine retention
- Overflow incontinence (Autonomous bladder)
 - Maximum capacity of urinary bladder → 500 ml
 - Beyond this → denervated muscle contracts
- Lax anal sphincter

CAUDA EQUINA SYNDROME

- Involvement of nerve roots - L1 to S5
- May present with U/L manifestations
- Presents with radicular pain/ root pain
- Knee jerk - L2, L3, L4 - absent
- Ankle jerk - S1, S2 - absent
- Asymmetrical areflexic paraplegia - LMN lesion
- Sensation loss - Saddle anesthesia
- Bladder/ Bowel involvement- Late presentation



- Subarachnoid space ends at S2
- Spinal cord ends at L1

Important Information

- Initial symptom/ complaint → lower back ache
 - D/t Mets to spine from Ca breast/ prostate

SACD (SUBACUTE COMBINED DEMYELINATION OF SPINAL CORD)

00:30:32

- B12 deficiency
- Dorsal column
 - Sensory ataxia
 - Loss of proprioception
 - (+) Romberg sign - eye closure imbalance
- Corticospinal pathway - (+) Babinski sign
- Peripheral nerves - Areflexia, Fasciculation
- Neurological features can exist in absence of Hematological features
 - Macrocytic anemia
 - Hypersegmented neutrophils
 - Reversible dementia
- Skin - hyperpigmentation of IP joints/ knuckles
- T/t - I/V vitamin B12



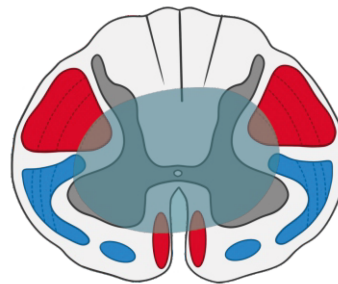
Important Information

- Romberg's sign
 - Eyes close → imbalance → SACD
 - Eyes open → imbalance → cerebellar pathology



CENTRAL CORD SYNDROME

00:36:02



Central cord syndrome
(Area affected)

CAUSES

Trauma - **hyperextension of neck**

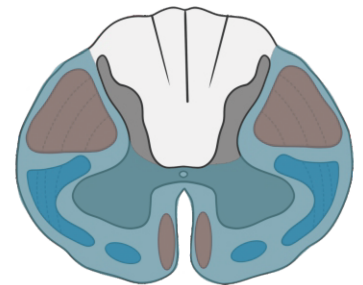
Syringomyelia

Spinal cord tumor
(meningioma, ependymoma)

- Tumor expansion Compression of corticospinal/ lateral spinothalamic pathway (cervical fibers)
- Corticospinal pathway
 - Arm weakness (spastic), **Brisk** DTR
- Lateral spinothalamic pathway
 - Dissociated sensory loss
 - Cape distribution sensory loss
- Sensations preserved
 - Light touch, Proprioception, Vibration

ANTERIOR CORD SYNDROME

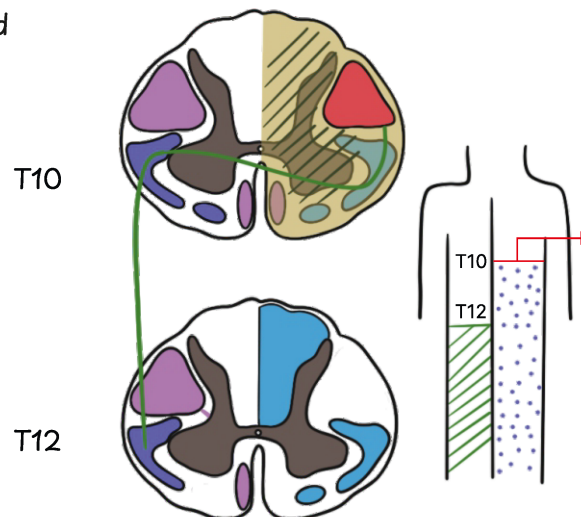
- Anterior spinal artery infarction (entire anterior spinal column affected)
- All motor and sensory and autonomic functions **lost**
- **Preserved proprioception, joint sensation, vibration**



BROWN SEQUARD SYNDROME

00:41:07

- Hemisection of spinal cord



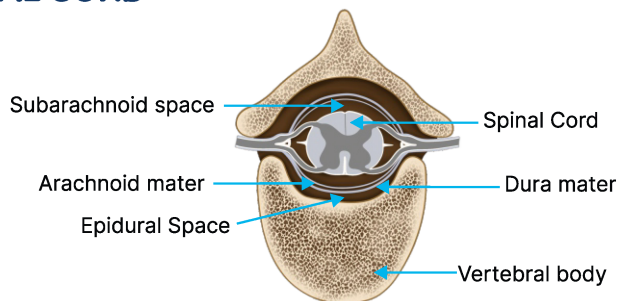
Free By - ZeroTrace



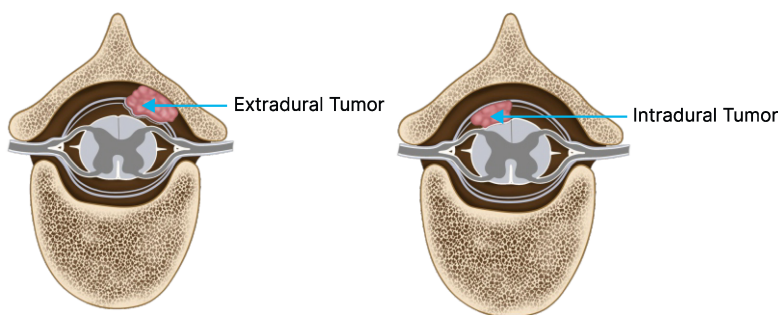
I/L PRESENTATION	C/L PRESENTATION
- Conscious proprioception loss - Vibration loss - Fine touch loss - Corticospinal tract involvement - T10 - radicular pain, fasciculation - Below T10 - I/L spasticity, I/L brisk DTR	Pain and temperature loss (2 level inferior)

COMPRESSION OF SPINAL CORD

00:44:54

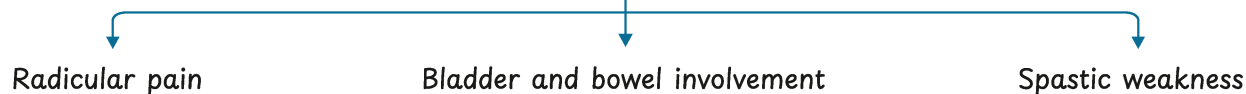


EXTRA AND INTRADURAL - EXTRAMEDULLARY COMPRESSION



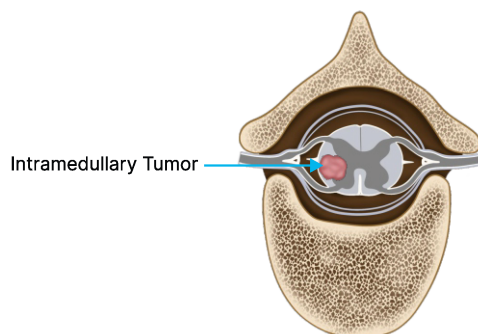
- Extradural tumor- Ca breast, Ca prostate
- Intradural tumor- Neurofibroma

FEATURES



INTRAMEDULLARY COMPRESSION

- Originates within spinal cord
 - Meningioma, Ependymoma, Astrocytoma
- Intramedullary tumor - sacral sparing





Q. A 55 year old male fell in bathroom today and came to ER. He has history of progressive weakness in his lower limbs and difficulty with urination and defecation. Work up shows elevated PSA levels. Examination shows saddle anesthesia with bilateral loss of ankle jerk reflexes and normal knee jerks. First differential diagnosis is?

- a. Cauda equina syndrome
- b. Conus medullaris syndrome
- c. Anterior cord syndrome
- d. Brown Sequard syndrome

Ans. (b)

Q. Which of the following is not correct about Subacute combined demyelination of spinal cord?

- a. Neurological features can coexist without hematological features
- b. Romberg sign is noted with eyes open
- c. Copper deficiency can mimic features of SACD
- d. Babinski sign with distal ataxia

Ans. (b)



13. CHANNELOPATHIES



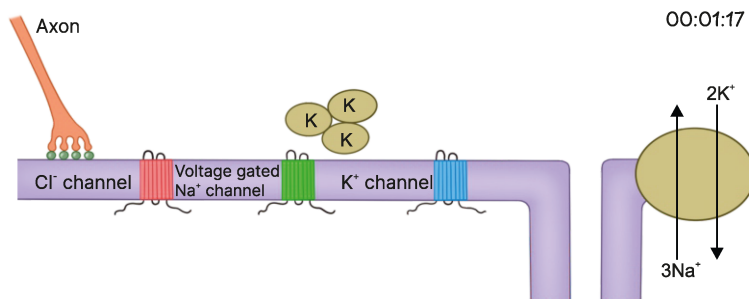
BECKER'S DISEASE

- Cl^- defect in skeletal muscles $\rightarrow \uparrow \text{Na}^+$ influx $\rightarrow$ repeated partial depolarization of muscle
- Myotonia - Defective/ slow relaxation of muscles



HYPOKALEMIC PERIODIC PARALYSIS

- Calcium channel defect
- Type I:
 - 90%
 - **CACNA1S gene** involved (Chromosome 1)
- Type II:
 - 10%
 - **SCN4A gene** involved (Chromosome 17)
 - **Na^+ channel defect**
 - **Loss of function** $\rightarrow$ failure to inactivate Na^+ / K^+ ATPase $\rightarrow \uparrow \text{K}^+$ influx and Na^+ efflux
- Age: 25 years
- Gender: Male
- Trigger: **High carbohydrate diet/ high salt diet/ prolonged exercise**
- Inheritance: Autosomal dominant
- Muscle involvement: **Proximal > distal muscles**
- Rationale:
 - Less K^+ makes the muscle membrane hyperpolarized and less excitable
- For type I, less calcium means less excitation-contraction coupling



CASE SCENARIO

- 25 A 25-year-old guy comes to the ER with recurrent episodes of muscle weakness in his legs in the morning hours. He says he did strenuous exercise in the gym last evening and had a large pizza for dinner. He is not able to raise his arms on command or get out of bed. GCS: 15/15. CN examination: normal. Flaccid weakness is noted in all limbs with proximal > distal weakness. $\text{K}^+ = 2.8 \text{ meq/L}$. He improves after potchlor administration. TFT: normal. Similar illness in sibling.

INFERENCE:

- Recurrent episodes of muscle weakness
- Large pizza $\rightarrow$ intake of carb/ salt
- Flaccid weakness: proximal > distal
- Hypokalemia (2.8 meq/L)
- Improvement with Potchlor solution
- Normal TFT $\rightarrow$ r/o Thyrotoxic periodic paralysis



Important Information

Hypokalemic periodic paralysis $\rightarrow$ $\uparrow$ sympathomimetic stimulation $\rightarrow$ K^+ influx worsens hypokalemic periodic paralysis

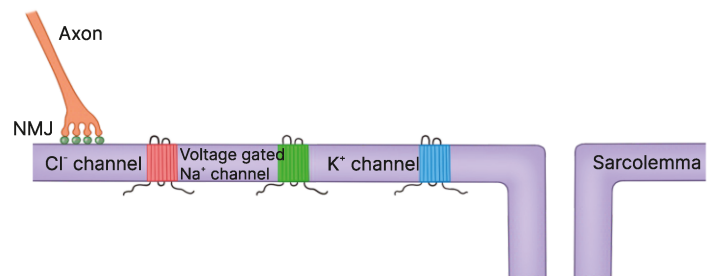
TREATMENT

- Mild attacks: no t/t
- Severe attacks: **KCl oral solution**
- Avoid carbs, salt, and strenuous exercise
- **Acetazolamide** improves type I and worsens type II hypokalemic periodic paralysis

HYPERKALEMIC PERIODIC PARALYSIS

00:08:44

- Gene involved: **SCN4A (Chromosome 17)**
- **Na^+ channel defect**
- **Gain of function** $\rightarrow$ Na^+ - K^+ pump fail to activate $\rightarrow$ persistent depolarization $\rightarrow$ $\downarrow$ K^+ influx
- Trigger: **K^+ rich diet**
- Inheritance: Autosomal dominant
- **Serum K^+** during attacks $\uparrow$
- Occurs in 1st decade of life



CASE SCENARIO

- A 10-year-old boy has severe muscle weakness after morning prayers in the school assembly. He was given bananas and coconut water in the medical room. Weakness worsened after it and was brought to ER. Reflexes are hypoactive with weakness in all limbs. Serum K^+ is elevated. Similar history in the father.

INFERENCE:

- Severe muscle weakness
- **K^+ rich diet** (banana/ coconut water) worsens the condition
- Reflexes: hypoactive
- Weakness in all limbs
- Serum **K^+ $\uparrow$**
- Same family history

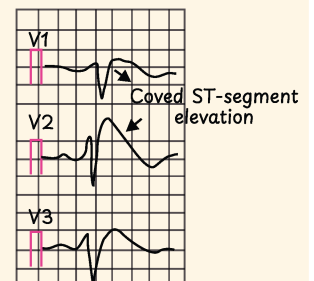
TREATMENT

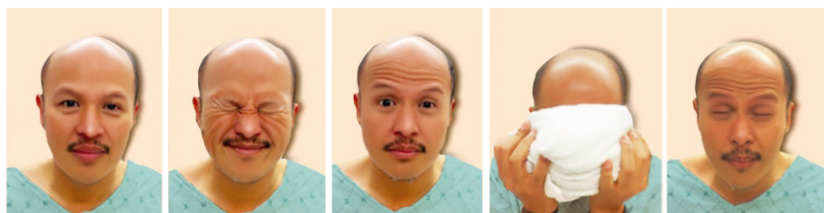
- **Acetazolamide**
 - Causes Kaliuria
- Avoid prolonged exercise

Important Information

BRUGADA SYNDROME

- SCN5A defect
- Na^+ channel defect
- Causes sudden nocturnal death
- **Coved ST-segment elevation**
- T/t: Implantable cardio defibrillator





At rest

Eyes closed
forcefully

Mild myotonia

Cold cloth on face

Myotonia worsens

- Allelic of Hyperkalemic PP
- Na^+ channel defect (SCN4A)
- Weakness after cold exposure, along with myotonia
- Myotonia: Delayed relaxation of muscles after contraction

CHANNEL DEFECTS

CALCIUM CHANNEL DEFECTS	POTASSIUM CHANNEL DEFECTS
• Lambert-Eaton syndrome • Familial hemiplegia migraine • Hypokalemic PP: Type 1	• Andersen-Tawil syndrome - Triad ○ Hypokalemic PP ○ Arrhythmias ○ Dysmorphic features (short stature, low-set ears)
• SKELETAL MUSCLE CHLORIDE CHANNEL DEFECT- Myotonia	
• Becker disease • Thomsen disease	



Q. Which of the following is a skeletal muscle chloride channel defect?

- a. Mucoviscidosis
- b. Becker disease
- c. Becker muscular dystrophy
- d. Hypokalemic periodic paralysis

Ans: (b)



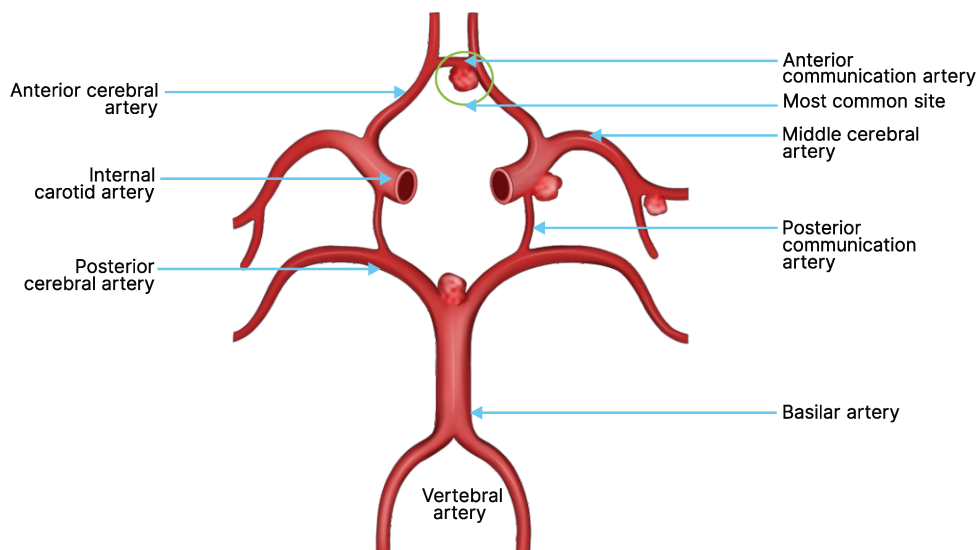
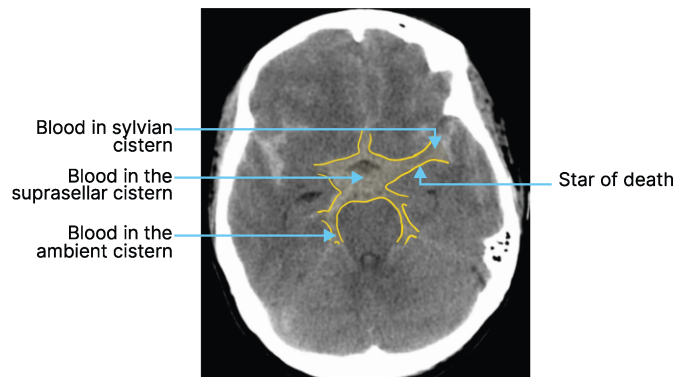
19. SUBARACHNOID HEMORRHAGE



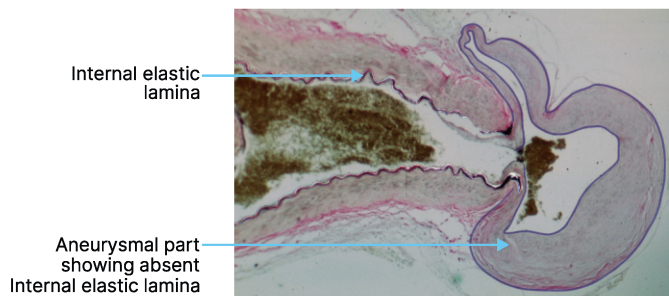
SUBARACHNOID HEMORRHAGE

00:00:11

- Accumulation of Blood in basal cistern
- One of causes of SAH → **Berry aneurysm**
- **M/c location of berry aneurysm** → Junction of Anterior communicating artery and Anterior cerebral artery
- If Ptosis is present → Aneurysm is at posterior communicating artery (aneurysm presses on oculomotor nerve → producing ptosis)



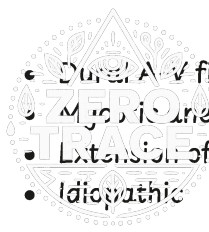
- (+) Nuchal rigidity → d/t blood causing **meningeal irritation**
- Degeneration of **internal elastic lamina**



CAUSES OF SAH

00:02:19

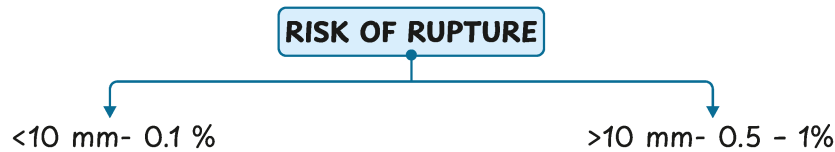
- M/c - **trauma**
- Rupture of Berry aneurysm (degeneration of internal elastic lamina layer)
- Arteriovenous malformation



- Dural A-V fistula
- Mycotic aneurysm – Complication of Acute bacterial endocarditis (*S. aureus*)
- Extension of intracerebral hemorrhage d/t hypertensive crisis
- Idiopathic

BASELINE DATA

- Mortality - 35 %
- Risk of rebleeding - 20%



- Giant Berry aneurysm - > 2.5 cm
- Location – Terminal internal carotid artery, middle cerebral artery bifurcation
- Mycotic aneurysm – *Staphylococcus aureus*
- M/c location - Anterior communicating artery
- M/c location with size and greatest risk of rupture -
 - > 8 mm, Posterior communicating artery (top of Basilar artery)
- Layer of Blood vessel defective in neck of berry aneurysm – Internal elastic lamina

CLINICAL FEATURES

CASE SCENARIO

25 year old female wakes up at 5 am with excruciating headache. She takes PCM tablet but headache worsens and she cant take it anymore and comes to ER at 8:30 am. On examination BP 160/100 mmHg, ptosis of left eye and nuchal rigidity noted. At 10:30 am she develops seizures and becomes stuporous.

INFERENCE

- ↑ headache even after medication , BP - ↑
- (+) ptosis, (+) nuchal rigidity
- Thunderclap headache (worst headache of my life)
- Short duration history
- Sudden headache + nuchal rigidity } SAH
- Intense painful condition
- Release of BNP → Natriuresis → salt loss in urine (Na < 125 meq) → seizures
- Seizures d/t dyselectrolytemia } Cerebral salt washing syndrome
 - Hypovolemic hyponatremia